Intersections of non-empty sets exist in Z

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openlogic:set-theory/z:local-snapshot-20261013; CC BY 4.0

Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel prop:intersectionsexist of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- the Separation axiom schema of Z
- the extensional description of the intersection as the set of elements shared by all members

Proposition 0.1. *If $A \neq \emptyset$, then $\bigcap A = \{ x \mid (\forall y \in A) x \in y \}$ exists.*

Proof. Let $A \neq \emptyset$, so there is some $c \in A$. Then $\bigcap A = \{ x \mid (\forall y \in A) x \in y \} = \{ x \in c \mid (\forall y \in A) x \in y \}$, which exists by Separation. □

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